

HW 6

#6.7 A) $f(a,b,c,d) = \sum m(0,3,4,5,7,9,11,13)$

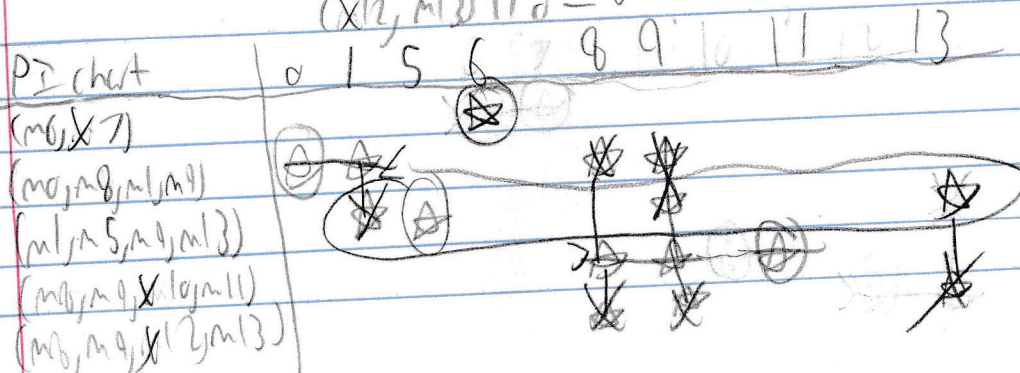
	CI	CII
G0	m0 0000 ✓	G0 (m0,m4) 0-00 PI
G1	m4 0100 ✓	G1 (m4,m5) 010- PI
G2	m3 0011 ✓	G2 (m3,m7) 0-11 PI
	m5 0101 ✓	(m3,m11) 0111 PI
	m9 1001 ✓	(m5,m7) 01-1 PI
G3	m7 0111 ✓	(m5,m13) 1011 PI
	m11 1011 ✓	(m9,m11) 10-1 PI
	m13 1101 ✓	(m9,m13) 1-01 PI

#6.9 B) $f(a,b,c,d) = \sum m(0,1,5,6,8,9,11,13) + \sum d(7,10,12)$

	CI	CII	CIII
G0	m0 0000 ✓	G0 (m0,m1) 000- ✓	G0 (m0,m6,m1,m9) -00- PI
G1	m1 0001 ✓	(m0,m8) -000 ✓	(m0,m1,m6,m9) -00-
	m8 1000 ✓	G1 (m1,m5) 0-01 ✓	G1 (m1,m5,m9,m13) -01 PI
G2	m5 0101 ✓	(m1,m9) -001 ✓	(m1,m9,m5,m13) -01
	m6 0110 ✓	(m6,m9) 100- ✓	(m6,m9,x10,m11) 10-- PI
	m9 1001 ✓	(m6,x10) 10-0 ✓	(m6,m9,x12,m13) 1-0- PI
	x10 1010 ✓	(m6,x12) 1-00 ✓	(m6,x10,m9,m11) 10--
	x12 1100 ✓	G2 (m5,x7) 01-1	(m6,x12,m9,m13) 1-0-
G3	x7 0111 ✓	(m5,m13) -101 ✓	
	m11 1011 ✓	(m6,x7) 011- ✓	
	m13 1101 ✓	(m9,m11) 10-1 ✓	
		(m9,m13) 1-01 ✓	
		(x10,m11) 101- ✓	
		(x12,m13) 110- ✓	

$f(a,b,c,d) = 011- + -00- + --01 + 10--$

$= ABC + BC + CD + AB$



$$\#6.14 \text{ A) } f(A,B,C,D) = \sum m(1,3,5,6,9,12,14,15) + \sum d(4,10,13)$$

C I

C II

C III

$$G1 \quad m1 \ 0001 \checkmark \quad G1(m1, m3) \ 00-1 \text{ PI}$$

$$x4 \ 0100 \checkmark \quad (m1, m5) \ 0-01 \checkmark$$

$$m6 \ 1000 \checkmark \quad (m1, m9) \ -001 \checkmark$$

$$G2 \quad m3 \ 0011 \checkmark \quad (x4, m5) \ 010- \text{PI}$$

$$m5 \ 0101 \checkmark \quad (x4, m6) \ 01-0 \checkmark$$

$$m6 \ 0110 \checkmark \quad (x4, m9) \ -100 \checkmark$$

$$m9 \ 1001 \checkmark \quad (m6, m9) \ 100- \text{PI}$$

$$x10 \ 1010 \checkmark \quad (m6, x10) \ 10-0 \checkmark$$

$$m12 \ 1100 \checkmark \quad (m6, m12) \ 1-00 \checkmark$$

$$G3 \quad x13 \ 1101 \checkmark \quad G2(m5, x13) \ -101 \checkmark$$

$$m14 \ 1110 \checkmark \quad (m6, m14) \ -110 \checkmark$$

$$G4 \quad m15 \ 1111 \checkmark \quad (m1, x13) \ 1-01 \checkmark$$

$$(x4, m14) \ 1-10 \checkmark$$

$$(m12, x13) \ 11-0 \checkmark$$

$$(m12, m14) \ 11-0 \checkmark$$

$$G3(x13, m15) \ 11-1 \checkmark$$

$$(m14, m15) \ 111- \text{PI}$$

$$G1(m1, m5, m9, x13) \ -01 \text{ PI}$$

$$\cancel{(m1, m9, m5, x13) \ -01}$$

$$(x4, m6, m12, x13) \ -1-0 \text{ PI}$$

$$\cancel{(x4, m6, m12, m14) \ -1-0}$$

$$(x4, m12, m5, x13) \ -10- \text{PI}$$

$$(x4, m12, m6, m14) \ -1-0 \text{ PI}$$

$$(m1, x10, m12, x13) \ 1-0- \text{PI}$$

$$\cancel{(m1, x10, m12, m14) \ 1-0-0}$$

$$(m6, m12, m9, x13) \ 1-0- \text{PI}$$

$$\cancel{(m6, m12, x10, m14) \ 1-0-0}$$

$$G2(m12, x13, x13, m15) \ 11-- \text{PI}$$

$$\cancel{(m12, m14, x13, m15) \ 11--}$$

PI chart | 1 3 5 6 9 12 14 15

(m1, m3)

(x4, m5)

(m6, m9)

(m14, m15)

(m1, m5, m9, x13)

(m4, m6, m12, x13)

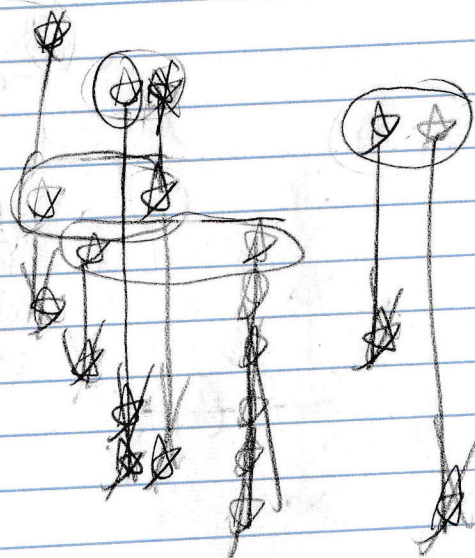
(x4, m12, m5, x13)

(x4, m12, m6, m14)

(m1, x10, m12, x13)

(m6, m12, m9, x13)

(m12, x13, x13, m15)



$$f(A,B,C,D) = 00-1 + 100- + 111- + --01 + -1-0$$

$$= \bar{A}\bar{B}D + ABC + ABC + (CD + \bar{C}\bar{D})$$

#6.18 A) $-0-1 = 3, 9, 11$ B) $\prod m(0, 4, 5, 6, 8, 12, 14)$
 $-01- = 2, 3, 11$ C) $\sum d(1, 10, 15)$
 $-11 = 3, 7, 11$ D) BCF COTAD
 $1--1 = 9, 11, 13$

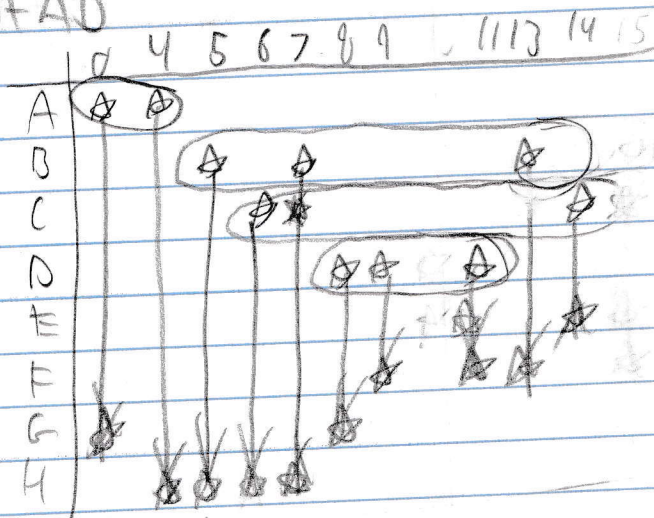
#6.21 A) $\prod m(1, 2, 3, 12)$

B) $A = 0000, 0100 \rightarrow \overline{w}y\overline{z}$

C) $0110, 0111, 1110, 1111 \rightarrow x\overline{y}$

D) $1000, 1001, 1010, 1011 \rightarrow w\overline{x}$

1) (A, B, C, D)



G.23 $F(A, B, C, D) = \sum m(7, 12, 14, 15) + \sum d(1, 3, 5, 9, 10, 11, 13)$

CI

G1 $x1(0011) \checkmark$
 $x9(1001) \checkmark$

G2 $x3(0011) \checkmark$
 $x5(0101) \checkmark$
 $x6(1010) \checkmark$
 $m12(1100) \checkmark$

G3 $m7(0111) \checkmark$
 $x11(1011) \checkmark$
 $x13(1101) \checkmark$
 $m14(1110) \checkmark$

G4 $m15(1111) \checkmark$

II

G1 $(x1, x3) 00-1 \checkmark$
 $(x1, x5) 0-01 \checkmark$
 $(x3, x6) 10-0 \checkmark$
 $(x3, m12) 1-00 \checkmark$

G2 $(x3, m7) 0-11 \checkmark$
 $(x3, x11) -011 \checkmark$
 $(x5, m7) 01-1 \checkmark$
 $(x5, x13) -101 \checkmark$
 $(x6, x11) 101- \checkmark$
 $(x6, m14) 1-10 \checkmark$
 $(m12, x13) 110- \checkmark$
 $(m12, m14) 11-0 \checkmark$

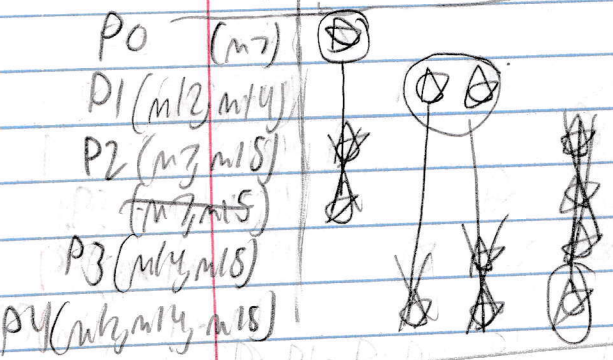
G3 $(m7, m15) -111 \checkmark$
 $(x11, m15) 1-11 \checkmark$
 $(x13, m15) 11-1 \checkmark$
 $(m14, m15) 11-- \checkmark$

III

G1 $(x1, x3, x5, m7) 0--1 \checkmark$ PI
 $(x3, x6, m12, m14) 1--0 \checkmark$ PI
 $(x3, m7, x6, m14) 1--0$

G2 $(x3, m7, x11, m15) --11 \checkmark$ PI
 $(x3, x11, m7, m15) --11$
 $(x5, m7, x13, m15) 1-1- \checkmark$ PI
 $(x5, x13, m7, m15) 1-1-$
 $(x6, x11, m14, m15) 1-1- \checkmark$ PI
 $(x6, m14, x11, m15) 1-1-$
 $(m12, x13, m14, m15) 11-- \checkmark$ PI
 $(m12, m14, x13, m15) 11--$

D_I 7 12 14 15



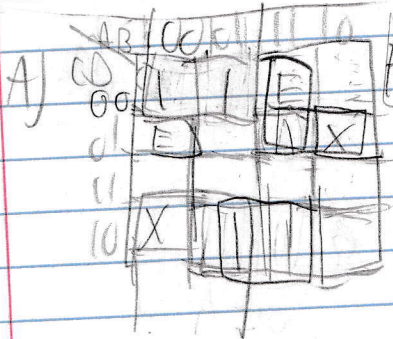
$$(P_0 + P_2)(P_1 + P_4)(P_1 + P_3 + P_4)(P_2 + P_3 + P_4)$$

$$(P_0 + P_2)(P_1 + P_4)(P_2 + P_3 + P_4)$$

$$(P_0 P_1 + P_0 P_4 + P_1 P_2 + P_2 P_4)(P_2 + P_3 + P_4)$$

$$F = P_0 P_1 P_2 + P_0 P_1 P_3 + P_0 P_1 P_4 + P_0 P_2 P_4 + P_0 P_3 P_4 + P_0 P_4 + P_1 P_2 + P_1 P_2 P_3 + P_1 P_2 P_4 + P_2 P_4 + P_2 P_3 P_4 + P_2 P_4$$

#6.24



$$F = \overline{A} \overline{D} + B \overline{C} \overline{D} + A \overline{C} \overline{D} + \overline{B} \overline{C} D E + A B C E$$